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Unit 3 : Tutorial 1 — De Morgan's Theorem & Design/Analysis of Basic and Advanced Adders

■ Part 1: De Morgan's Theorem

- Theorem 1: The complement of a sum of variables equals the product of their complements: $(A + B)' = A' \cdot B'$
- Theorem 2: The complement of a product of variables equals the sum of their complements: $(A \cdot B)' = A' + B'$
- Equivalent circuits: Theorem 1 — a NOR gate equals an AND gate with inverted inputs (bubbled AND). Theorem 2 — a NAND gate equals an OR gate with inverted inputs (bubbled OR).
- Simplifications: a) $(A + B)' = A' \cdot B'$ b) $(X \cdot Y)' = X' + Y'$ c) $(A' + B' + C')' = A \cdot B \cdot C$ d) $((A + B) \cdot C)' = (A + B)' + C' = A'B' + C'$
- Proof of Theorem 1 by truth table: rows (A,B) = 00/01/10/11 give $(A+B)' = 1,0,0,0$ and $A' \cdot B' = 1,0,0,0$ — identical columns, hence proved.
- Chemical process alarm: let H = heater, F = fan, P = pump. Process active = H.F.P; alarm X = (H.F.P)'. By De Morgan: $X = H' + F' + P'$ — the alarm sounds if the heater OR fan OR pump is off; implement with an OR gate with inverted inputs.

■ Part 2: Design and Analysis of Basic and Advanced Adders

- Half adder truth table: (A,B) 00→S0 C0; 01→S1 C0; 10→S1 C0; 11→S0 C1. Sum = $A'B + AB' = A \oplus B$; Carry = $A \cdot B$. Logic diagram: one XOR gate for Sum, one AND gate for Carry.
- Full adder truth table: the 8 rows give $S = A \oplus B \oplus C_{in}$ and $C_{out} = 1$ whenever two or more inputs are 1. SOP forms: Sum = $A'B'C_{in} + A'BC_{in}' + AB'C_{in}' + ABC_{in}$; Cout = $A'BC_{in} + AB'C_{in} + ABC_{in}' + ABC_{in} = AB + BC_{in} + AC_{in}$.
- Full adder from two half adders + OR gate: HA1 adds A and B; HA2 adds HA1's sum with C_{in} ; the two carry outputs go to an OR gate producing C_{out} .
- 4-bit ripple-carry adder: four full adders cascaded, the carry-out of each feeding the carry-in of the next higher stage.
- Main disadvantage: propagation delay — each stage waits for the previous carry, so the delay ripples from LSB to MSB. With 10 ns per full adder, total delay = $4 \times 10 \text{ ns} = 40 \text{ ns}$.
- Adding A = 1101 and B = 1011: bit0 $1+1+0 \rightarrow S0=0 \text{ c}=1$; bit1 $0+1+1 \rightarrow S1=0 \text{ c}=1$; bit2 $1+0+1 \rightarrow S2=0 \text{ c}=1$; bit3 $1+1+1 \rightarrow S3=1 \text{ c}=1$. Sum = 10000 (5-bit result, $13 + 11 = 24$), showing the ripple effect.
- Carry-lookahead adder: computes the carry of every stage directly from the inputs instead of waiting for the ripple. Advantage: much higher speed because all carries are generated in parallel.
- Generate $G_i = A_i \cdot B_i$ (carry created when both inputs are 1). Propagate $P_i = A_i \oplus B_i$ (carry passed on when exactly one input is 1).
- Comparison — Ripple-carry: slow (delay $\propto$ number of bits), small area, low complexity. Carry-lookahead: fast (delay grows logarithmically), larger area, high complexity.
- A = 1011, B = 0110 ripple addition: Bit0: $A0=1, B0=0, C_{in}=0 \rightarrow S0=1, c=0$; Bit1: $A1=1, B1=1, C_{in}=0 \rightarrow S1=0, c=1$; Bit2: $A2=0, B2=1, C_{in}=1 \rightarrow S2=0, c=1$; Bit3: $A3=1, B3=0, C_{in}=1 \rightarrow S3=0, c=1$. Final Sum = 0001, Carry-out = 1.